

Mend-Amar Badral

 MendeBadra |  Mend-Amar Badral |  mendebadra.github.io
 mendamar.badral@gmail.com |  +36 70 7385 744 |  Budapest, Hungary

Driven by an open mindset with a strong drive for learning new technologies, I aspire to contribute to developing genuine solutions for real world, impactful problems.

EDUCATION

Budapest University of Technology and Economics (BME) Sep, 2025 – June, 2027
M.Sc. in Computer Science Engineering
Budapest, Hungary
Specialization: Data Science & Artificial Intelligence

National University of Mongolia (NUM) Sep, 2021 – June, 2025
B.Sc. in Applied Mathematics
Ulaanbaatar, Mongolia
GPA: 3.2/4.0 (rank 3/34)
Supervisor: [Galtbayar Artbazar](#)
Thesis: [Evaluating land degradation using image processing methods](#)

EXPERIENCE

Center of Mathematics Application Research Lab Oct, 2023 – June, 2025
Undergraduate Lab Lead & Research Intern

- Selected as undergraduate Lab Lead; participated in weekly seminars and collaborative discussions with faculty members on machine learning, statistical modeling, and applied image analysis.
- Developed a custom vendor-specific **Python pipeline** for processing multispectral drone imagery and sensor metadata to generate vegetation and land degradation maps.
- Worked extensively with multispectral image data, vegetation indices (NDVI, MSAVI), raster data processing, and geospatial analysis using Python tools including NumPy, OpenCV, Rasterio, and GDAL.
- Implemented feature engineering and threshold optimization approaches for interpretable land classification and environmental analysis from UAV imagery.

PROJECTS

[Data Integration for 3D Gaussian Splatting](#)

- Implemented a custom JSON parser to extract sensor data obtained by aiSim, an autonomous driving simulator by aiMotive, into a [nerfstudio](#) (3D reconstruction software) format. Performed LiDAR point cloud sensor fusion with the camera images to obtain color information.

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Implemented polynomial regression from Bishop & Bishop's *Deep Learning Foundations and Concepts*, and reproduced key theoretical results in interactive Julia notebooks to empirically analyze **overfitting**, **underfitting**, and the **bias-variance tradeoff**.

AWARDS

Stipendium Hungaricum Scholarship Sep, 2025

SKILLS

Languages: English (C1 [IELTS 8.0](#))

Programming: Python, SQL, Bash, JavaScript, Julia, C/C++, R

Tools: Git, Docker, pandas, xarray, sklearn, rasterio, PyTorch, GDAL, OpenCV, nerfstudio

Data Engineering: Large-scale data processing, multi-threading, relational databases (PostgreSQL/MySQL), file formats (GeoTIFF, Parquet, HDF5)